realized, we have a phenomenal, short-lived, outward burst. In other words, we have exactly what the big bang theory was missing: a *bang*, and a big one at that. That's why Guth's discovery is something to get excited about.[12]

The cosmological picture emerging from Guth's breakthrough is thus the following. A long time ago, when the universe was enormously dense, its energy was carried by a Higgs field perched at a value far from the lowest point on its potential energy bowl. To distinguish this particular Higgs field from others (such as the electroweak Higgs field responsible for giving mass to the familiar particle species, or the Higgs field that arises in grand unified theories[13]) it is usually called the inflaton field.[25] Because of its negative pressure, the inflaton field generated a gigantic gravitational repulsion that drove every region of space to rush away from every other; in Guth's language, the inflaton drove the universe to *inflate*. The repulsion lasted only about 10^{-35} seconds, but it was so powerful that even in that brief moment the universe swelled by a huge factor. Depending on details such as the precise shape of the inflaton field's potential energy, the universe could easily have expanded by a factor of 10^{30}, 10^{50}, or 10^{100} or more.

These numbers are staggering. An expansion factor of 10^{30}—a conservative estimate—would be like scaling up a molecule of DNA to roughly the size of the Milky Way galaxy, and in a time interval that's much shorter than a billionth of a billionth of a billionth of the blink of an eye. By comparison, even this conservative expansion factor is billions and billions of times the expansion that would have occurred according to the standard big bang theory during the same time interval, and it exceeds the total expansion factor that has cumulatively occurred over the subsequent 14 billion years! In the many models of inflation in which the calculated expansion factor is much larger than 10^{30}, the resulting spatial expanse is so enormous that the region we are able to see, even with the most powerful telescope possible, is but a tiny fraction of the whole universe. According to these models, none of the light emitted from the vast majority of the universe could have reached us yet, and much of it won't arrive until long after the sun and earth have died out. If the entire cosmos were scaled down to the size of earth, the part accessible to us would be much smaller than a grain of sand.